

Homework 5

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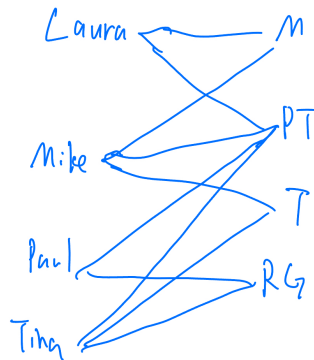
1 Problem 1

(a)

A matching of G is the following:

Connect 1 with a, Connect 2 with d, Connect 3 with b, Connect 4 with c. No matching exist in H , because the set $\{1, 2, 4\}$'s neighbor set is $\{b, c\}$, which has a smaller size. Therefore, according to Hall's theorem, no such matching exists in H .

(b)



There exist a matching:

Laura-M

Mike-PT

Paul-RG

Tina-T

Therefore, Hall's condition must be satisfied according to Hall's theorem. And a assignment can be:

Laura-M

Mike-PT

Paul-RG

Tina-T

(c)

This problem can transition to a matching problem. The set A is the professors set, and set B is the committees set. Each professor's vertex has 3 neighbors. And each committees set has 6 neighbors. We need to prove there exists a matching covering set B.

We can use hall's theorem to verify it. Suppose the set S has n vertices. Then, the total degree is 6n. Then, $N(S)$ has at least 6n degrees in total. Because each professor only has 3 neighbors, there must exist 2n professors. Because 2n is greater than or equals to n, hall's condition is verified, and there exists a matching covering set B, and we are done.

2 Problem 2

(a)

Lemma: any k-regular bipartite graph contains a perfect matching:

Since it is a K-regular bipartite graph we can divide the vertices in to set A and B base on their color. Therefore, the total degree in A is $K * |A|$ which equals to the total degree in B, which is $K * |B|$. Therefore, $|A|=|B|$. For any sets S in A, the total degree is $K * |S|$. Therefore, the total degree of $N(S)$ must be greater than or equals to $K * |S|$. Then, the total number of vertices is greater than or equals to $|S|$. Then, hall's condition is satisfied. The lemma is proven.

Back to the problem. Since it is a k-regular graph, it has a perfect matching. Then we can delete all edges in the matching to form a edge set. And the graph become a k-1-regular graph, and it also has a perfect matching. We can repeat this process for k times. Then, we got k edge sets. They are disjoint and are perfect matching. Therefore, its edge set E can be written as the union of k edge-disjoint perfect matchings.

(b)

The total degree in A and B is same. Therefore,

$$|A|a = |B|b$$

Since,

$$|A| \leq |B|$$

Therefore,

$$a \geq b$$

For any edge set S in A. The total degree is $|S|a$. And the total degree of its neighbor set must greater than or equals to its total degree. Therefore,

$$|N(S)|b \geq |S|a$$

Since,

$$a \geq b$$

Then

$$|N(S)| \geq |S|$$

And Hall's condition is satisfied, then there exists a matching covering A. Then, G contains a matching of size $|A|$.

3 Problem 3

(a)

We can pick any edge between A and B. And let be a part of a matching. Since the end points of the edge is matched, we can delete them and all edge connecting them. For the rest of the set, any non-empty set X in A (without the deleted vertex), $|N(X)| > |X|$ before the deleting. If the deleted vertex in B is in $N(X)$, then new $|N(X)|$ will decrease 1, so $|N(X)| \geq |X|$. And hall's condition is meet. Therefore, there exists a matching covering rest of A. Add back the deleted matching. We found a matching covering A with one specific edge to include. Therefore, every edge of G belongs to some matching M that covers A.

(b)

We can first prove that there exists a matching of size $|A| - d_*$, we can add d_* additional vertices to set B, and let them connect to all vertices in A. Then $|N(S)|$ increases d_* for all S. Since d_* is the maximum difference of $|N(S)|$ and $|S|$, then hall's condition is satisfied. Therefore, A has a matching covering A with $B + d_*$ additional vertices. Then we can delete these d_* additional vertices, and the matching size will decrease at most d_* . Then, there still exists a matching of size $|A| - d_*$.

Then, we can prove why this is the maximum.

Because there exist a set S that $|S| - |N_G(S)| = d_*$. In original graph. The set S can only match vertices in $N_G(S)$. Therefore, d_* vertices cannot be matched. Since only in set S d_* vertices cannot be matched, in set A, greater than or equal to d_* vertices cannot be matched.

Therefore, the maximum matching of G has size $|A| - d_*$.

(c)

Given a set $S \in A$ that violates Hall's condition, consider the set of vertices H in B not adjacent to S. All H in won't have neighbor in set S.

Because

$$|S| > |N(S)|$$

Then because $|A| = |B|$

$$|A/S| < |B/N(S)| = |H|$$

And because

$$|N(H)| \leq |A/S|$$

Therefore,

$$|N(H)| < |H|$$

Then, H also violates Hall's condition. And we are done.